

SWIDWIN, Poland

WMO: 122120

Lat: 53.791N

Lon: 15.826E

Elev: 385

StdP: 14.49

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.7	13.6	2.2	6.2	11.0	7.6	8.2	16.2	32.4	43.7	29.0	41.0	5.7	100	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	84.3	66.8	80.7	65.4	77.6	64.0	69.2	80.4	67.4	77.3	65.7	74.0	9.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	95.5	72.8	64.0	90.4	70.8	62.4	85.6	69.7	33.5	80.6	32.1	77.1	30.7	74.2	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.6	20.5	18.0	DB	3.5	90.2	7.2	3.7	-1.7	92.9	-5.9	95.1	-10.0	97.2	-15.2	99.9	
			WB	3.9	72.4	7.5	2.1	-1.5	73.9	-5.8	75.2	-10.0	76.4	-15.4	77.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	29.5	31.7	37.7	47.5	54.7	60.8	64.8	64.3	56.7	47.8	39.8	34.1
	DBStd	14.08	8.89	7.86	7.81	7.51	7.24	6.08	5.77	5.34	5.48	6.33	5.95	7.78
	HDD50	2618	634	512	384	133	29	1	0	0	7	119	306	493
	HDD65	6543	1099	932	846	524	329	156	76	79	255	534	755	958
	CDD50	1728	0	0	3	60	176	326	460	443	208	50	2	0
	CDD65	176	0	0	0	0	11	32	70	57	6	0	0	0
	CDH74	1593	0	0	0	8	129	308	680	422	46	0	0	0
	CDH80	382	0	0	0	0	20	67	194	97	4	0	0	0
Wind	WSAvg	8.5	9.8	9.2	8.9	8.6	7.7	7.8	7.7	7.5	7.8	8.4	9.2	9.9
Precipitation	PrecAvg	26.4	2.0	1.7	1.9	1.4	2.2	2.6	3.2	2.9	2.3	1.9	2.0	2.3
	PrecMax	35.6	4.6	3.6	3.9	3.9	3.9	5.4	6.5	6.5	5.7	4.5	4.7	4.0
	PrecMin	20.8	0.1	0.3	0.4	0.5	0.8	0.8	0.5	0.9	0.5	0.9	0.1	0.5
	PrecStd	3.6	1.2	0.7	0.9	0.7	0.8	1.0	1.7	1.2	1.0	1.0	0.9	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.3	51.0	62.3	74.7	83.0	86.7	89.8	88.3	80.2	67.7	55.4	52.0
		MCWB	47.9	46.0	50.0	58.5	64.4	69.5	68.7	69.2	66.2	58.8	51.9	49.6
	2%	DB	46.2	47.4	57.2	69.9	77.7	81.2	85.2	82.3	74.0	63.8	51.9	48.6
		MCWB	44.2	43.6	46.8	55.0	62.3	65.3	67.1	66.3	61.4	57.0	49.6	46.7
	5%	DB	43.5	44.3	52.5	65.6	72.8	77.4	81.2	78.4	70.4	60.2	50.0	46.1
		MCWB	41.8	41.1	45.6	52.5	59.3	63.5	65.6	64.4	60.0	55.1	48.0	44.1
	10%	DB	41.0	41.9	49.0	61.3	68.8	73.2	77.5	75.2	66.6	57.0	48.2	43.6
		MCWB	39.6	39.2	43.9	50.0	57.2	61.4	64.6	63.7	58.4	53.4	46.5	41.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.8	48.2	52.4	60.4	66.8	71.9	71.0	71.0	67.2	60.9	52.1	50.2
		MCDB	50.4	49.7	59.1	73.2	78.2	83.9	82.7	83.8	78.0	65.3	54.0	51.6
	2%	WB	44.4	44.3	49.2	56.2	63.8	67.8	69.1	68.9	63.3	58.0	49.9	47.1
		MCDB	46.0	46.4	53.8	66.9	74.8	77.8	81.4	78.8	70.2	61.8	51.4	48.4
	5%	WB	42.0	41.5	46.6	53.6	61.4	65.2	67.5	66.7	61.3	55.8	48.3	44.2
		MCDB	43.2	43.8	51.3	63.4	70.5	74.1	78.2	74.7	67.3	59.4	49.6	45.5
	10%	WB	39.7	39.3	44.1	51.1	58.7	62.9	65.7	65.1	59.8	53.7	46.5	42.0
		MCDB	40.9	41.7	48.4	59.5	65.9	70.8	74.3	71.8	65.2	56.6	47.9	43.3
Mean Daily Temperature Range	5% DB	MDBR	7.8	9.7	13.8	18.3	18.9	18.3	17.8	17.3	15.6	12.5	8.0	7.2
		MCDBR	7.7	12.1	20.3	24.6	24.2	24.3	24.6	22.5	21.0	16.2	10.2	8.8
		MCWBR	7.1	9.3	12.3	13.1	11.7	12.2	10.1	10.5	10.5	10.3	8.3	8.0
	5% WB	MCDBR	7.7	10.9	17.8	22.9	22.0	21.8	21.6	19.2	17.8	14.8	9.6	8.6
	MCWBR	7.4	8.9	12.0	12.8	11.4	12.0	10.3	10.2	10.1	10.2	8.1	8.1	
Clear-Sky Solar Irradiance	taub	0.307	0.333	0.364	0.403	0.401	0.397	0.420	0.417	0.377	0.358	0.336	0.305	
	taud	2.297	2.300	2.273	2.231	2.286	2.322	2.274	2.290	2.384	2.400	2.357	2.305	
	Ebn at Noon	206	239	256	261	268	271	261	254	251	228	193	182	
	Edh at Noon	19	26	33	39	39	38	39	36	30	23	18	16	
All-Sky Solar Radiation	RadAvg	172	383	771	1320	1663	1755	1620	1369	993	515	207	120	
	RadStd	21	55	91	172	191	138	203	123	109	75	29	12	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (34 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	-254	+112	N/A	

Nomenclature: See separate page